

CONTACT

Location: Lille, France (National & International Mobility)

Phone: +33 6 05 73 14 16

Email: hanzsintim@gmail.com

LinkedIn: linkedin.com/in/hanzofosuhene-sintim

GitHub: github.com/HanzSintim

Portfolio: hanzsintim.github.io

EDUCATION

2025 – Present (Expected 2028)
JUNIA ISEN — Lille, France
Master's Degree in Engineering (M1)
Specialisation: Electronics & Embedded Systems

2023 – June 2027 (In progress)
KNUST — Kumasi, Ghana
BSc in Computer Engineering
4th Year completed concurrently with JUNIA ISEN Master's

SKILLS

Low-Level & Programming
Embedded C, C, C++17, Assembly (MSP430), Python 3, SQL

Embedded Systems & Hardware
STM32, ESP32, MSP430, Arduino Uno, UART/PWM, Sensors, Semiconductor Physics

Testing, Analysis & Tools
Hardware-Software Debugging (GDB), Signal Chain Validation, Interface Testing, Git/Linux, KiCad

Interpersonal (Soft Skills)
Team Leadership, Analytical Problem-Solving, Rigor, Autonomy, Cross-Cultural Communication

LANGUAGES

• **English: C2 (Native Speaker)** — Ghanaian by birth; full schooling & BSc completed in English in Ghana.

• **French: A2 (Elementary)** — Daily & academic immersion in Lille, France (Duolingo Score: 21).

INTERESTS & ACTIVITIES

• **History & Science:** Reading world history and exploring how physical/engineering systems work.

• **Community Leadership:** Organizing Lead at GHAFRA Nord (Lille).

HANZ OFOSUHENE-SINTIM

ENGINEERING STUDENT — ELECTRONICS & EMBEDDED SYSTEMS

Seeking an Internship in Embedded C / Assembly Development, Testing & Analysis
Duration: 3 to 4 Months | Availability: April – July / August 2027

PROFILE

M1 Engineering student in **Electronics and Embedded Systems** at JUNIA ISEN Lille, concurrently completing my 4th-year BSc in Computer Engineering at KNUST (Ghana). Skilled in low-level programming (**Embedded C, C/C++17, Assembly**), microcontroller firmware (**STM32, ESP32, MSP430**), and systematic **hardware-software testing, debugging, and analysis**. Rigorous and autonomous, with proven executive leadership representing 800+ engineering students, I am seeking a **3 to 4-month internship starting in April 2027** to contribute to embedded firmware development, testing, and system validation.

TECHNICAL PROJECTS

Autonomous Street-Cleaning Robot — JUNIA ISEN Lille **2026 – Present (Ongoing)**
M1 Engineering R&D Project (7-Member Team) · Embedded C, STM32, ESP32, UART

- **Developed** real-time **Embedded C** firmware on an **STM32** microcontroller for PWM H-bridge motor and brush actuation, paired with **ESP32** wireless telemetry.
- **Conducted** hardware-software integration testing, **UART** serial frame analysis (AI camera), and closed-loop control verification.

Digital Thermometer & Signal Acquisition Chain **2025**
Embedded Measurement System · MSP430 Microcontroller, C & Assembly

- **Designed and programmed** low-level firmware in **C and Assembly** on a TI **MSP430** microcontroller for precision temperature measurement.
- **Analyzed and validated** the complete signal processing chain from analog sensor capture and ADC conversion to calibrated digital readout.

Automated Water Level Control System **2024**
Embedded Control & Sensor Calibration · Arduino Uno (C/C++), Ultrasonic Sensors

- **Programmed** deterministic microcontroller logic for automatic pump actuation governed by **calibrated level thresholds**.
- **Performed** sensor accuracy analysis and threshold hysteresis testing to prevent relay chatter and ensure reliable fluid regulation.

E.Leclerc Self-Checkout Terminal & Sales Architecture (C++17) **2024 – 2026**
Software Architecture & Validation Testing · C++17, OOP, Custom Data Structures

- **Architected** a zero-dependency **C++17** transaction engine evolving an online sales server architecture into an interactive self-checkout terminal.
- **Executed** boundary and unit testing on runtime polymorphic (`dynamic_cast`) load-cell weight verification and a 12-denomination Euro change algorithm.

PROFESSIONAL EXPERIENCE & LEADERSHIP

Department President — ACES, KNUST **Aug 2025 – May 2026 (10 mos)**
Association of Computer Engineering Students (Higher Education) · Executive Board · Kumasi, Ghana

- **Directed** executive governance, budget management, and multidisciplinary teams representing **800+ Computer Engineering students**.
- **Spearheaded** technical and logistical execution of **CODEFEST 2026**, coordinating industry sponsors, speakers, and technical judging panels.

Software Development & Testing Intern — Spagard Technologies **Oct – Dec 2023 (3 mos)**
IT & Software Services Sector · Backend & Quality Assurance Unit · Ghana

- **Developed** server-side software components for internal enterprise management systems within a structured engineering workflow.
- **Structured** relational databases and **executed** systematic code debugging, root-cause analysis, and interface testing prior to release.